

Every nonzero injective tensor factor descends the Bishop-Phelps-Bollobas property

Run `fa_banach.001`, lane 5

2026-06-19

Status

This packet gives a full qualitative answer to the tensor-factor descent question following Proposition 2.8 in Aron, Choi, Kim, Lee, and Martin [1]. The source proves the result for $C(K)$ and asks what other spaces have the same descent property. The theorem below shows that, as a yes/no BPBp statement, every nonzero Banach tensor factor E works.

There is a modulus nuance. The source's $C(K)$ argument preserves the same modulus $\eta(\varepsilon)$. The all- E theorem below gives the modulus $\varepsilon \mapsto \eta(\varepsilon/2)$. A same-modulus refinement still holds under the stronger state-norming-unit, or spear-vector, hypothesis discussed at the end.

Source question

The relevant passage appears on page 11 of the source paper, immediately after Proposition 2.8.

not have even Lindenstrauss property B [44].

A more general way of stating Proposition 2.8 is that if $(X, C(K) \hat{\otimes}_\varepsilon Y)$ has the BPBp with $\eta(\varepsilon)$, then so does (X, Y) . We do not know what other spaces, besides $C(K)$, have this property. In an analogous way, noting that $L_1(\mu, X) = L_1(\mu) \hat{\otimes}_\pi X$, we remark that we do not know if a result similar to Proposition 2.8 can be obtained for vector valued L_1 -spaces in the domain: that is, we do not know if the fact that $(L_1(\mu, X), Y)$ has the BPBp implies that (X, Y) does as well (where X and Y are Banach spaces and μ is a positive measure).

Transcription of the part used here: Proposition 2.8 may be restated as saying that if $(X, C(K) \hat{\otimes}_\varepsilon Y)$ has the BPBp with $\eta(\varepsilon)$, then so does (X, Y) . The authors write that they do not know what other spaces, besides $C(K)$, have this property.

Definitions

All Banach spaces are over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. A pair (X, Y) has the Bishop-Phelps-Bollobas property with function η if, whenever $T \in S_{\mathcal{L}(X, Y)}$ and $x_0 \in S_X$ satisfy

$$\|Tx_0\| > 1 - \eta(\varepsilon),$$

there are $S \in \mathcal{L}(X, Y)$ and $x_1 \in S_X$ such that

$$\|S\| = \|Sx_1\| = 1, \quad \|S - T\| < \varepsilon, \quad \|x_1 - x_0\| < \varepsilon.$$

Idea of the proof

The key point is that no special topology of $C(K)$ is needed to find a norming evaluation. For any Banach space E , the injective tensor product $E \widehat{\otimes}_\varepsilon Y$ embeds canonically into $C(B_{E^*}, Y)$, where B_{E^*} carries the weak-star topology. Hence every tensor element is a continuous Y -valued function on a compact space and attains its norm somewhere.

Choose any $e \in S_E$ and embed Y by $y \mapsto e \otimes y$. If the BPBp in $E \widehat{\otimes}_\varepsilon Y$ gives a norm-attaining approximant $\tilde{S}$ close to $e \otimes T$, evaluate $\tilde{S}x_1$ at a norming functional $e_1^* \in B_{E^*}$. The scalar $\alpha = e_1^*(e)$ must be close to the unit circle because $\tilde{S}x_1$ is close to $e \otimes Tx_1$. Evaluation at e_1^* , followed by a scalar rotation, therefore yields an exactly norm-attaining Y -valued operator close to T .

Theorem 1. *Let E be a nonzero Banach space. Let X and Y be Banach spaces. If the pair $(X, E \widehat{\otimes}_\varepsilon Y)$ has the BPBp, then the pair (X, Y) has the BPBp.*

More precisely, if $(X, E \widehat{\otimes}_\varepsilon Y)$ has the BPBp with function η , then (X, Y) has the BPBp with function

$$\varepsilon \mapsto \eta(\varepsilon/2).$$

Proof. Fix $e \in S_E$. We use the canonical isometric representation

$$E \widehat{\otimes}_\varepsilon Y \hookrightarrow C(B_{E^*}, Y),$$

where B_{E^*} is weak-star compact and an elementary tensor $a \otimes y$ is represented by the function

$$e^* \mapsto e^*(a)y.$$

For finite tensors this is exactly the injective norm formula; the completed tensor product is the corresponding closed subspace of $C(B_{E^*}, Y)$.

Let $0 < \varepsilon < 1$ and put $\rho = \varepsilon/2$. Suppose $T \in S_{\mathcal{L}(X, Y)}$ and $x_0 \in S_X$ satisfy

$$\|Tx_0\| > 1 - \eta(\rho).$$

Define $J : Y \rightarrow E \widehat{\otimes}_\varepsilon Y$ by $J(y) = e \otimes y$, and set $\tilde{T} = J \circ T$. Since $\|e\| = 1$, J is an isometry, so $\tilde{T} \in S_{\mathcal{L}(X, E \widehat{\otimes}_\varepsilon Y)}$ and

$$\|\tilde{T}x_0\| = \|Tx_0\| > 1 - \eta(\rho).$$

Applying the BPBp for $(X, E \widehat{\otimes}_\varepsilon Y)$ with tolerance ρ , we get $\tilde{S} \in \mathcal{L}(X, E \widehat{\otimes}_\varepsilon Y)$ and $x_1 \in S_X$ such that

$$\|\tilde{S}\| = \|\tilde{S}x_1\| = 1, \quad \|\tilde{S} - \tilde{T}\| < \rho, \quad \|x_1 - x_0\| < \rho.$$

As an element of $C(B_{E^*}, Y)$, the function $\tilde{S}x_1$ attains its norm. Choose $e_1^* \in B_{E^*}$ such that

$$\|(\tilde{S}x_1)(e_1^*)\| = 1.$$

Let $\alpha = e_1^*(e)$. Since $\tilde{T}x_1(e_1^*) = \alpha Tx_1$, we have

$$1 = \|(\tilde{S}x_1)(e_1^*)\| \leq \|(\tilde{S}x_1 - \tilde{T}x_1)(e_1^*)\| + |\alpha| \|Tx_1\| < \rho + |\alpha|.$$

Thus $|\alpha| > 1 - \rho > 0$. Choose $\omega \in \mathbb{K}$ with $|\omega| = 1$ and $\omega\alpha = |\alpha|$.

Define

$$S = \omega (e_1^* \otimes \text{Id}_Y) \circ \tilde{S} \in \mathcal{L}(X, Y).$$

Then $\|S\| \leq 1$, while

$$\|Sx_1\| = \|(\tilde{S}x_1)(e_1^*)\| = 1.$$

Hence $\|S\| = \|Sx_1\| = 1$. Finally,

$$\begin{aligned}\|S - T\| &\leq \|\omega(e_1^* \otimes \text{Id}_Y)(\tilde{S} - \tilde{T})\| + \|\omega(e_1^* \otimes \text{Id}_Y)\tilde{T} - T\| \\ &\leq \rho + \|\omega\alpha T - T\| = \rho + (1 - |\alpha|) < 2\rho = \varepsilon.\end{aligned}$$

Also $\|x_1 - x_0\| < \rho < \varepsilon$. Therefore (X, Y) has the BPBp with modulus $\varepsilon \mapsto \eta(\varepsilon/2)$. $\square$

Same-modulus refinement

The proof above loses a factor of two because the norming functional e_1^* need only satisfy $|e_1^*(e)| > 1 - \rho$, not $e_1^*(e) = 1$. If E has a norm-one element e whose state face

$$K_e = \{e^* \in B_{E^*} : e^*(e) = 1\}$$

norms E , then the earlier argument over K_e gives the same modulus $\eta(\varepsilon)$. This is the state-norming-unit condition, equivalent to the usual spear-vector formulation [3]; it includes closed unital subspaces of $C(K)$ as a special case.

Scope and limitations

This is a full answer for the qualitative BPBp descent problem associated with injective tensor factors: every nonzero E works. It does not prove that the same original modulus $\eta(\varepsilon)$ is preserved for every E . The same modulus is proved here only under the additional state-norming-unit condition.

The separate $L_1(\mu, X)$ domain question in the same source paragraph is not addressed by this packet.

Packet history

This full packet supersedes two intermediate packets now preserved in the local **history/** folder. The first proved the same-modulus result for closed unital subspaces of $C(K)$. The second replaced that concrete hypothesis by the abstract state-norming-unit, or spear-vector, condition. Both are now historical review aids; the ledger and run registry point to the present full packet.

Novelty check

The run indexes contained no exact packet or attempt for arXiv:1305.6420 and no entry for the all-nonzero injective tensor factor formulation. A bounded web search for phrases including “BPBp”, “injective tensor”, “ $C(K, Y)$ ”, “unital uniform algebra”, and “spear vector” found the source paper, the spear-vector monograph [3], and a related later preprint by Grando and Santos [2], which proves a $C_0(L, Y)$ descent theorem and related generalized AHSP stability results. The downloaded arXiv source of that later preprint did not state the all-nonzero injective tensor theorem proved here.

Human-review recommendation

The key verifier focus is the canonical isometric embedding $E \hat{\otimes}_\varepsilon Y \subset C(B_{E^*}, Y)$ and the scalar-rotation estimate turning evaluation at e_1^* into a norm-attaining operator within ε of T . The proof is otherwise a direct quantitative refinement of the source paper’s Proposition 2.8 argument.

References

- [1] R. Aron, Y. S. Choi, S. K. Kim, H. J. Lee, and M. Martin, *The Bishop-Phelps-Bollobas version of Lindenstrauss properties A and B*, Trans. Amer. Math. Soc. 367 (2015), 6085–6101; arXiv:1305.6420.
- [2] T. Grando and E. R. Santos, *Stability results of the Bishop-Phelps-Bollobas property and the generalized AHSP*, arXiv:2506.17762.
- [3] V. Kadets, M. Martin, J. Meri, and A. Perez, *Spear operators between Banach spaces*, arXiv:1701.02977.